

GAZE: Game-Adaptive Zone of Engagement

Notes:

- [] Record and embed timeframe in method & setup

Diagram ideas:

Input : face detection(emotions) -> open AI create game (based on emotions) -> robot speaks and interact (verbal , hand and face gestures.)

- if frustrated -> hints.
- If angry -> easier game
- If happy-> celebrate and smile and give harder games
- If Bored -> speeds up
- If seem upset -> cheer them up
- ...

Operation of a Cognitive Robot

Cognitive robots achieve their goals by

- **Perceiving** their environment
 - Paying **attention** to the events that matter
 - **Anticipating** the need for some action
 - **Planning** what to do
 - **Anticipating** the outcome as it executes the action
 - **Learning** from the resultant interaction
 - **Adapting** to change
- []

Robot idea:

- [] integrate ws-10 for facial recognition and emotion detection
- [] Generates games with openai on the fly
- [] Facial recognition (Happy? frustrated?) if frustrated give hints if happy celebrate via /ws-10
- [] If bored speeds it up
- [] If angry - no worries lets try another game (easier)
- [] If answers are correct, it gives harder puzzles.
- [] Make easier if too hard
- [] If sad games to cheer him up
- [] Camera - detects face (emotions) - game logic - open ai generates response - robot speaks and reacts.
- [] Robot personality - funny , sarcastic , serious.
- [] Games - puzzles, riddles, trivia, memory games, word games, math games, etc.
- [] Use openai to generate games and responses based on the user's emotions and performance.
- [] timer to track how long the user has been playing and adjust the game difficulty accordingly.

- [] scoring system to track the user's progress and provide feedback on their performance.
 - [] reward system to encourage the user to keep playing and improving their skills.
 - [] feedback system to allow the user to provide feedback on the games to change robot's considerations and personality.
 - [] feature to allow the user to customise the robot's personality and game preferences.
 - [] feature to allow the user to save their progress and continue playing later.
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Task Allocation

report-related:

- **Salman:** Introduction (10% of the task mark)
- **Alfie:** Background (10% of the task mark)
- **Alfie:** Method and setup (35% of the task mark)
- **Salman:** Results/outcome/or system analysis (30% of the task mark)
- **Alfie: Conclusion** (10% of the task mark)
- **Anyone:** References (5% of the task mark)

Code-related:

Salman: making prompts

Salman: gaming logic

Alfie: facial recognition

Alfie: general code architecture

Alfie

- Code
- Talk about results
- Write conclusion

Salman

- Plan up
 - Testing
 - Small tasks
 - Write introduction
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Task (4):

Introduction (S) (<220-word)

...

Background (A) (<220-word)

...

Method & Setup (A & S) (<810-word) (

...

Results/outcome/or system analysis (S & A) (<730-word)

...

Conclusion (A) (<220-word)

...

References

...

Ai Declaration stuff:

...

Plan 2: story-telling robot

- Bored - adds action
- Scared - toned
- Happy - continues

Plan 3: social-skills trainer:

- Detects eye contact
- Detect emotions
- Detect gestures (closed off?)
- Gives feedback based on the above things